

Entity Binding Mechanisms Across Positional Embedding and Hybrid Architectures*

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Abstract

A key capability of language models is *entity binding*—associating entities with their attributes for later retrieval. Recent work has shown that models using RoPE positional embeddings implement this through three mechanisms: positional, lexical, and reflexive. However, it remains unknown how alternative positional encodings and architectures that combine attention with state-space (Mamba) layers affect these mechanisms. We investigate entity binding across ALiBi, hybrid Mamba-Attention, and pure Mamba (SSM) architectures, extending beyond the RoPE models studied in prior work. Using interchange interventions (activation patching), we trace how binding information flows through model layers and, for hybrid architectures, isolate the contributions of individual components. Our experiments reveal that: (1) ALiBi models exhibit the same three binding mechanisms with the same progression pattern as RoPE models; (2) in hybrid models (Falcon-H1, Zamba2), the attention component is the primary carrier of binding information, while the Mamba component contributes negligibly; (3) binding transitions in Zamba2 occur exclusively at hybrid (attention-containing) layers; and (4) pure Mamba models achieve entity binding but with a different, less structured signal—reaching only $\sim 50\%$ reflexive binding compared to $\sim 90\%$ in attention-based models, with no intermediate positional or lexical phases. These findings suggest that attention is superior to state-space layers for entity binding: it produces stronger, more structured binding signals regardless of positional encoding, and remains the dominant binding mechanism even when combined with Mamba layers in hybrid architectures.

1 Introduction

In-context reasoning requires language models to bind entities to their attributes and retrieve them later. For example, given “*Ann loves pie*”, a model must bind *Ann* to *pie* to answer “*Who loves pie?*”. This *entity binding* capability is a prerequisite for many downstream tasks, from coreference resolution to multi-hop question answering.

Gur-Arieh et al. (2025) showed that Transformer models with RoPE positional embeddings implement entity binding through three mechanisms: **positional** (binding via position in the context), **lexical** (binding via token identity), and **reflexive** (binding via the entity-attribute pairing itself). These mechanisms depend on both layer depth and entity position in the context, with reflexive binding becoming dominant at later layers. However, their analysis was limited to RoPE-based Transformer models, leaving open the question of how other architectures implement entity binding.

This question is particularly relevant given the growing diversity of language model architectures. Other architectures use different positional encodings, such as ALiBi (Press et al., 2022). Beyond positional encoding, state-space models (SSMs) like Mamba (Gu and Dao, 2023) offer a fundamentally different approach to sequence modeling. Recent *hybrid* architectures combine attention with Mamba layers within a single model (Zuo et al., 2025; Zypfra, 2024), raising the question of which component carries binding information.

In this work, we extend the entity binding analysis beyond RoPE to three additional architecture families. We make the following contributions:

- We replicate and extend the interchange intervention framework of Gur-Arieh et al. (2025) to ALiBi models (MPT-7B, Bloomz-3B), confirming that the positional, lexical, and reflexive binding mechanisms observed in RoPE models generalize to alternative positional

*Code available at <https://github.com/avivyossef29/mixing-mechs-nlp.git> and <https://github.com/NitzanZacharia/100rep.git>

encodings. Furthermore, localized patch-effect analysis reveals that both models follow a similar binding trajectory: transitioning from an early phase of suppressed intervention effects featuring a transient positional bias—an expected artifact of ALiBi’s distance penalty—through an intermediate phase of concurrent mechanism activation, before ultimately converging into dominant reflexive binding alongside a persistent residual lexical signal.

- We provide the first analysis of entity binding in hybrid Mamba-Attention models (Falcon-H1-3B, Zamba2-2.7B). Using *multi-layer component patching*—which isolates cumulative component contributions by patching a single component across all layers simultaneously—we demonstrate that **attention is the primary carrier of entity binding**: patching the attention component at all layers produces 92% reflexive binding in Falcon-H1, while patching the Mamba component yields 99.5% no effect. In Zamba2, binding transitions occur exclusively at hybrid (attention-containing) layers.
- We provide the first analysis of entity binding in a pure Mamba model (Falcon-Mamba-7B), revealing a qualitatively different binding pattern: a weaker reflexive signal, no intermediate positional or lexical phases, and a primacy-recency effect absent in attention-based models.
- We extend the task to 100 bound entities to assess the scalability of these binding mechanisms. Attention-based ALiBi models demonstrate clear adaptability, shifting their routing computations earlier in the network to redistribute representational effort across layers. Pure Mamba and hybrid architectures, by contrast, exhibit catastrophic state saturation. Despite attaining near-perfect accuracy at 20 entities, they collapse uniformly across the sequence, falling below 10% accuracy at scale.

2 Background and Related Work

2.1 Entity Binding in Language Models

Entity binding refers to the ability of language models to associate entities with their attributes within context. Feng and Steinhardt (2024) showed that

language models attach *binding ID* vectors to corresponding entities and attributes, forming a continuous subspace. Dai et al. (2024) further demonstrated that these binding representations encode the *ordering* of entities—essentially a positional mechanism. However, Gur-Arieh et al. (2025) showed that this positional mechanism generalizes poorly as the number of bound entities increases, and identified two additional mechanisms—lexical and reflexive—that compensate for positional binding’s limitations.

The experimental framework uses *interchange interventions* (Geiger et al., 2021): given a base input and a counterfactual source input, hidden activations from the source run are patched into the base run at specific layers. The resulting change in model output reveals which mechanism the model relies on at that layer. Three counterfactual types isolate the three mechanisms: a *positional* counterfactual places a different entity at the same position, a *lexical* counterfactual places the same entity at a different position, and a *reflexive* counterfactual changes both position and identity, testing whether the model tracks the entity-attribute pairing directly.

2.2 Positional Encoding Methods

Language models encode token position in diverse ways. **RoPE** (Su et al., 2021) applies rotation matrices to query and key vectors, encoding relative position through the angle between rotated vectors. **ALiBi** (Press et al., 2022) adds a linear bias to attention scores proportional to the distance between tokens, avoiding explicit positional embeddings entirely. **Absolute learned embeddings** add a trainable vector per position to the input, as used in GPT-2 (Radford et al., 2019) and OPT (Zhang et al., 2022). These encoding methods provide different inductive biases for how models represent token position.

2.3 State-Space Models and Hybrid Architectures

State-space models (SSMs), particularly Mamba (Gu and Dao, 2023) and its successor Mamba-2 (Dao and Gu, 2024), offer an alternative to attention by modeling sequences through a recurrent state that is updated at each token. Unlike attention, Mamba does not compute pairwise token interactions, raising the question of whether it can implement the same binding mechanisms. Pure Mamba models such as Falcon-Mamba-7B

Architecture	Model	Acc.
RoPE	Gemma-2-2B-IT [†]	100%
ALiBi	MPT-7B-Instruct	100%
	Bloomz-3B	100%
Pure SSM	Falcon-Mamba-7B	100%
Hybrid	Falcon-H1-3B-Instruct	100%
	Zamba2-2.7B-Instruct-v2	100%

[†] RoPE baseline from Gur-Arieh et al. (2025).

Table 1: Models and entity binding accuracy (20 groups).

(Zuo et al., 2024) demonstrate that SSMs alone can achieve competitive language modeling performance.

Recent hybrid architectures combine both paradigms (Lieber et al., 2025). **Falcon-H1** (Zuo et al., 2025) uses a parallel design where each layer computes Mamba, attention, and MLP outputs simultaneously, and their outputs are combined and added to the residual stream. **Zamba2** (Zyphra, 2024) uses a sequential design with 54 layers: most are pure Mamba layers, with 9 hybrid layers (at positions 6, 12, 18, 24, 30, 36, 42, 47, 51) where the input first passes through a shared attention block and then through a Mamba block—notably, attention output is not added directly to the residual stream but instead serves as input to the Mamba layer. Zamba2 shares only 2 attention blocks across all 9 hybrid layers.

3 Method

3.1 Task and Data Generation

We adopt the entity binding task from Gur-Arieh et al. (2025). Each input consists of n entity-attribute pairs (e.g., “*the lamp is in Box A. the book is in Box B. ...*”) followed by a query (e.g., “*Which box contains the lamp?*”). The model must retrieve the correct attribute. We use $n = 20$ entity-attribute pairs (referred to as *entity groups*), ensuring the task is non-trivial and requires robust binding. Table 1 lists the models used in our experiments. All models achieve 100% accuracy on this task, confirming they can perform entity binding reliably.

For each base input, we construct a counterfactual input by rearranging the entity-attribute pairs. We then patch the hidden state from the counterfactual run into the base run and observe which answer the model produces. Following Gur-Arieh et al. (2025), consider:

Base: the lamp is in Box A, the book is in Box B,

the cup is in Box C, ... *Which box contains the cup?* **Answer: Box C**

Counterfactual: the cup is in Box A, the lamp is in Box B, the book is in Box C, ... *Which box contains the lamp?* **Answer: Box B**

After patching the counterfactual’s hidden state into the base run, the model’s output reveals which mechanism it relies on:

- **Positional:** the model outputs Box C (the box at the cup’s original position in the counterfactual—position 3 holds “the book” in Box C)—it binds by position.
- **Lexical:** the model outputs Box A (the box paired with “cup” in the counterfactual)—it binds by token identity.
- **Reflexive:** the model outputs Box B (the counterfactual’s answer)—it tracks the query’s entity-attribute pairing directly.

3.2 Residual Stream Patching

To identify *where* binding mechanisms emerge, we perform per-layer activation patching on the residual stream. For each layer ℓ , we replace the hidden state at the last input token (where the model generates its answer) with the corresponding hidden state from the counterfactual run. The intervention effect is measured by whether the model’s output changes to match the counterfactual prediction. If patching at layer ℓ changes the output, it means binding-relevant information has entered the residual stream by that layer. By sweeping over layers, we trace how binding mechanisms emerge across the network. For each patched sample, we classify the intervention effect as one of: *no effect* (output unchanged), *positional* (output matches positional counterfactual prediction), *lexical* (matches lexical counterfactual), *reflexive* (matches reflexive counterfactual), or *mixed* (output changes but does not match any single counterfactual prediction).

3.3 Component Patching

For hybrid models, we isolate the contribution of individual components (attention vs. Mamba) by patching the output of a specific component rather than the full residual stream. The interpretation depends on architecture: in Falcon-H1, where components are computed in parallel and their outputs are scaled and added to the residual stream, patching a component changes its direct contribution to the residual. In Zamba2, where attention output

feeds sequentially into the Mamba block, patching attention changes the input that the Mamba block receives, whose output is then added to the residual stream.

We patch a component at all layers simultaneously—for Falcon-H1 this means all layers, and for Zamba2 only the 9 hybrid layers that contain attention. This replaces the component’s output with the corresponding output from the counterfactual run across the entire forward pass, while leaving all other components unmodified. Formally, for a component c and layer set $\mathcal{L}$:

$$\forall \ell \in \mathcal{L} : \quad \mathbf{o}_\ell^c \leftarrow \mathbf{o}_\ell^{c,cf} \quad (1)$$

where $\mathbf{o}_\ell^c$ is the output of component c at layer ℓ .

4 Results

4.1 Residual Stream Patching Across Architectures

We perform residual stream patching on all models using 1,000 samples per layer. Figure 2 shows the layer-wise distribution of binding mechanisms across all architectures.

ALiBi models. MPT-7B-Instruct (32 layers) exhibits a clear transition through all three binding mechanisms. Layers 0–16 show no intervention effect ($\sim 94\%$ no effect). At layers 17–20, positional and lexical binding emerge (peaking at 12% and 22% respectively at layer 20). Layer 21 marks a sharp transition to reflexive binding (44%), which then dominates through the final layers, reaching 89% at layer 31.

Bloomz-3B (30 layers) shows the same pattern but with a sharper, later transition. Layers 0–21 are flat ($\sim 99\%$ no effect). Reflexive binding jumps from 39% at layer 25 to 89% at layer 26, reaching 94% at the final layer. Unlike MPT-7B, Bloomz-3B shows almost no positional phase (never exceeding 1%). Both ALiBi models also exhibit entity-position dependence: first and last entities are bound positionally, while middle entities transition through lexical to reflexive binding (Figure 3b), matching the pattern observed in RoPE models by Gur-Arieh et al. (2025).

Hybrid models. Falcon-H1-3B (32 layers) shows a characteristic transition: layers 0–16 show no effect, layers 17–21 are dominated by lexical

binding, and layers 22–31 are dominated by reflexive binding (up to 92% at later layers). Falcon-H1 shows the same entity-position dependence as attention-only models (Figure 3c), confirming that the hybrid architecture preserves attention’s binding structure.

Zamba2-2.7B reveals a striking pattern: binding transitions occur *exclusively* at hybrid layers (Figure 4). The mechanism remains flat across pure Mamba layers, with abrupt transitions at hybrid layer positions (e.g., layer 30 $\rightarrow$ 31 shows a jump to positional binding; layer 36 $\rightarrow$ 37 jumps to 91% reflexive). This provides strong evidence that attention is responsible for binding transitions.

Pure Mamba. Falcon-Mamba-7B (64 layers), despite achieving 100% task accuracy, shows a fundamentally different binding pattern. Layers 0–41 are completely flat ($\sim 55\%$ no effect, $\sim 37\%$ mixed). Reflexive binding begins emerging around layer 46 (7%) and accelerates in the final layers, reaching 50.5% at layer 63. Notably, positional and lexical binding are nearly absent throughout (never exceeding 5%), and the mixed category remains high ($\sim 40\%$) even at the final layer. This contrasts sharply with attention-based models, which reach $\sim 90\%$ reflexive with clear intermediate phases.

Furthermore, Falcon-Mamba-7B exhibits a *primacy-recency effect* (Figure 3d): binding signal is strongest for entities at the beginning and end of the context, while middle entities show weaker effects. This is consistent with the limitations of recurrent state representations, where middle-position information is partially overwritten. In contrast, all attention-based architectures—RoPE, ALiBi, and Hybrid—show the same rich entity-position-dependent structure (Figure 3a–c): the first entity group is bound positionally, middle groups transition through lexical to reflexive binding, consistent with the findings of Gur-Arieh et al. (2025). Mamba lacks this structured position dependence entirely.

4.2 Scaling to 100 entities.

We perform residual stream patching on all models at 100 bound entities, using 1,000 samples per layer. While the 20-entity setting establishes the structure of binding mechanisms across architectures, it does not assess robustness under load. Scaling to 100 entities allows us to probe binding capacity under pressure and verify that weak or inconsistent signals observed at 20 entities reflect genuine

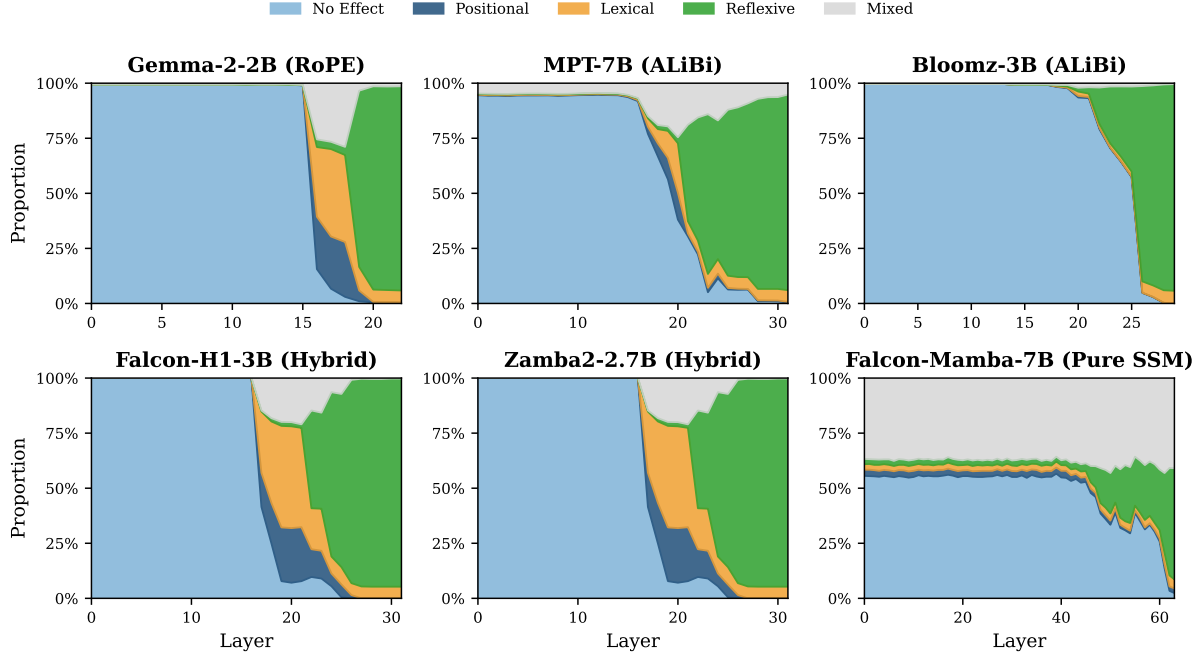


Figure 1: Residual stream patching results across architectures. Top row: Gemma-2-2B (RoPE baseline from Gur-Arieh et al. (2025)) and ALiBi models show the same clean transition to reflexive binding. Bottom row: hybrid models show architecture-dependent transitions, and pure Mamba shows a different, later transition with persistent mixed effects.

mechanisms rather than random noise.

ALiBi models. When scaled to 100 bound entities, MPT-7B-Instruct shows severely degraded binding, consistent with its lower task accuracy (61.9%). Layers 0–14 lack a true dormant phase, sustaining $\sim 61\%$ no-effect alongside a persistent $\sim 38\%$ mixed signal. Positional and lexical emergence is heavily suppressed (peaking at just 2.3% and 7.0%), replaced by a prolonged mixed phase peaking at 77.9% (layer 21). Reflexive binding rises slowly, reaching only 61.6% at the final layer.

Conversely, Bloomz-3B largely preserves its original pattern. Layers 0–13 remain flat ($\sim 92\%$ no effect), followed by a protracted mixed phase through layer 25, after which reflexive binding jumps sharply from 46% to 80% and reaches 93% at the final layer. Positional and lexical phases remain negligible.

However, the entity-position dependence observed at 20 entities is substantially diminished: While a slight positional bias persists for the very first entities in the sequence, the recency effect is largely absent, with the final entities no longer relying on positional binding.

Architecture	Model	Accuracy (%)
RoPE	Gemma-2-2B	8.80
ALiBi	MPT-7B-Instruct	61.90
	Bloomz-3B	96.27
Hybrid	Falcon-H1-3B-Instruct	8.30
	Zamba2-2.7B	6.80
Pure SSM	Falcon-Mamba-7B	0.80

Table 2: Model accuracy on the entity binding task scaled to 100 entity groups.

Hybrid Models. **Falcon-H1-3B** collapses to 8.3% reflexive accuracy at 100 entities, yet its layer-wise distribution preserves a recognizable structural echo of its 20-entity behavior. Layers 0-16 are dominated by mixed ($\sim 75\%$ with a residual no-effect component ($\sim 17\%$), and the characteristic transition remains visible at layers 17–22: no-effect collapses, mixed shifts, and reflexive begins to grow - reaching 15.8% at the final layer. The transition architecture is preserved; only its magnitude is reduced. Crucially, no fallback strategy emerges: positional, lexical, and no-effect rates remain indistinguishable from baseline noise throughout, confirming that the model does not degrade gracefully by reverting to simpler mechanisms.

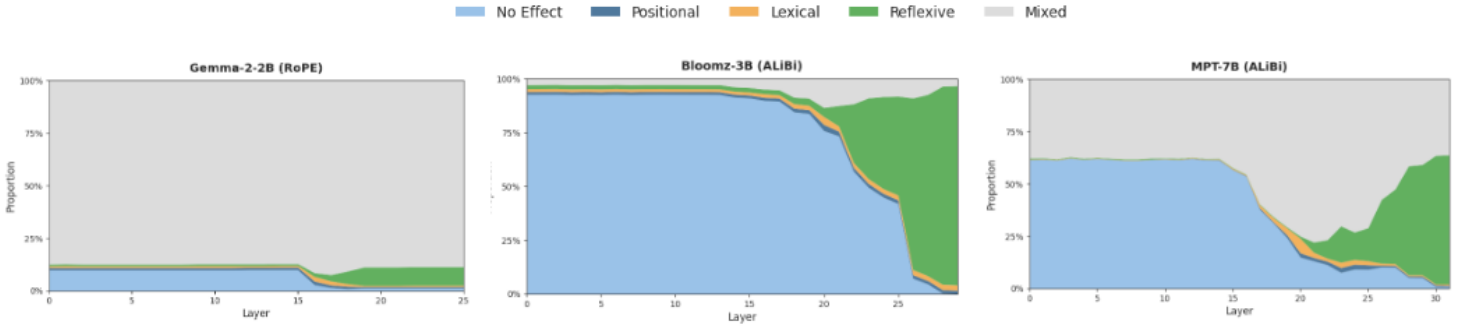


Figure 2: Residual stream patching results at 100 entity groups for RoPE and ALiBi models.

Zamba2.. At 100 entities the model’s progression is almost entirely suppressed — in stark contrast to the complete phase transition observed at 20 entities. The model holds flat at $\sim 89.6\%$ mixed and $\sim 7.5\%$ no-effect through layer 35, with reflexive never exceeding 0.5% , compared to near 90% reflexive binding at the same layer under 20 entity load. At layer 36 (a hybrid layer), reflexive jumps to 1.3% , then 6.8% at layer 37, plateauing there through layer 53. The timing is preserved; the cascade is not. What remains is a single residual step where a full phase transition once occurred.

Pure Mamba. Falcon-Mamba-7B collapses entirely at 100 entities, yielding just 0.8% reflexive accuracy. All 64 layers show a uniform 96% mixed signal with every other mechanism flatlined near 1% — no transition, no dormant phase, no layer at which binding information enters the residual stream. This is qualitatively different from the other collapsed models, which retain at least a faint structural echo of their binding machinery.

4.3 Component Patching in Hybrid Models

We perform component patching using 200 samples per experiment.

Falcon-H1. Patching either attention or Mamba at any *single* layer produces 100% no effect, because each component’s per-layer contribution is too small relative to the residual stream. However, patching across all layers simultaneously reveals a clear picture (Table 3): attention carries the binding information (92% reflexive), while Mamba contributes negligibly (99.5% no effect). Not all attention layers contribute equally: patching the

Component	No Eff.	Pos.	Lex.	Refl.	Mixed
Attention comp. (all layers)	0%	0%	8%	92%	0%
Mamba comp. (all layers)	99.5%	0%	0%	0%	0.5%

Table 3: Multi-layer component patching results for Falcon-H1-3B (200 samples). Patching the attention component at all layers produces 92% reflexive binding; patching the Mamba component has virtually no effect.

attention component at layers 17–19 produces zero effect, while layers 24–30 alone yield 68.5% reflexive binding. This confirms that the binding signal is concentrated in later attention layers, consistent with the residual stream results.

Zamba2. The Zamba2 architecture provides a natural control experiment. With only 9 hybrid layers out of 54, most of the model consists of pure Mamba layers. The residual stream results show that binding transitions occur exclusively at hybrid (attention-containing) layers (Figure 4), while pure Mamba layers produce no change in binding mechanism. In Zamba2’s sequential design, attention output feeds into the Mamba block rather than being added directly to the residual stream. Patching attention at any single hybrid layer produces 96.5% no effect, indicating that no individual layer is sufficient. Patching attention at all hybrid layers yields 90% reflexive binding—closely matching Falcon-H1’s 92% . Note that in Zamba2, this result reflects the attention-mediated pathway (attention followed by Mamba), rather than attention alone, since attention output feeds into the Mamba block.

5 Discussion and Conclusion

Attention appears to drive entity binding in hybrid models. Our results suggest that attention, rather than Mamba, is the primary carrier of en-

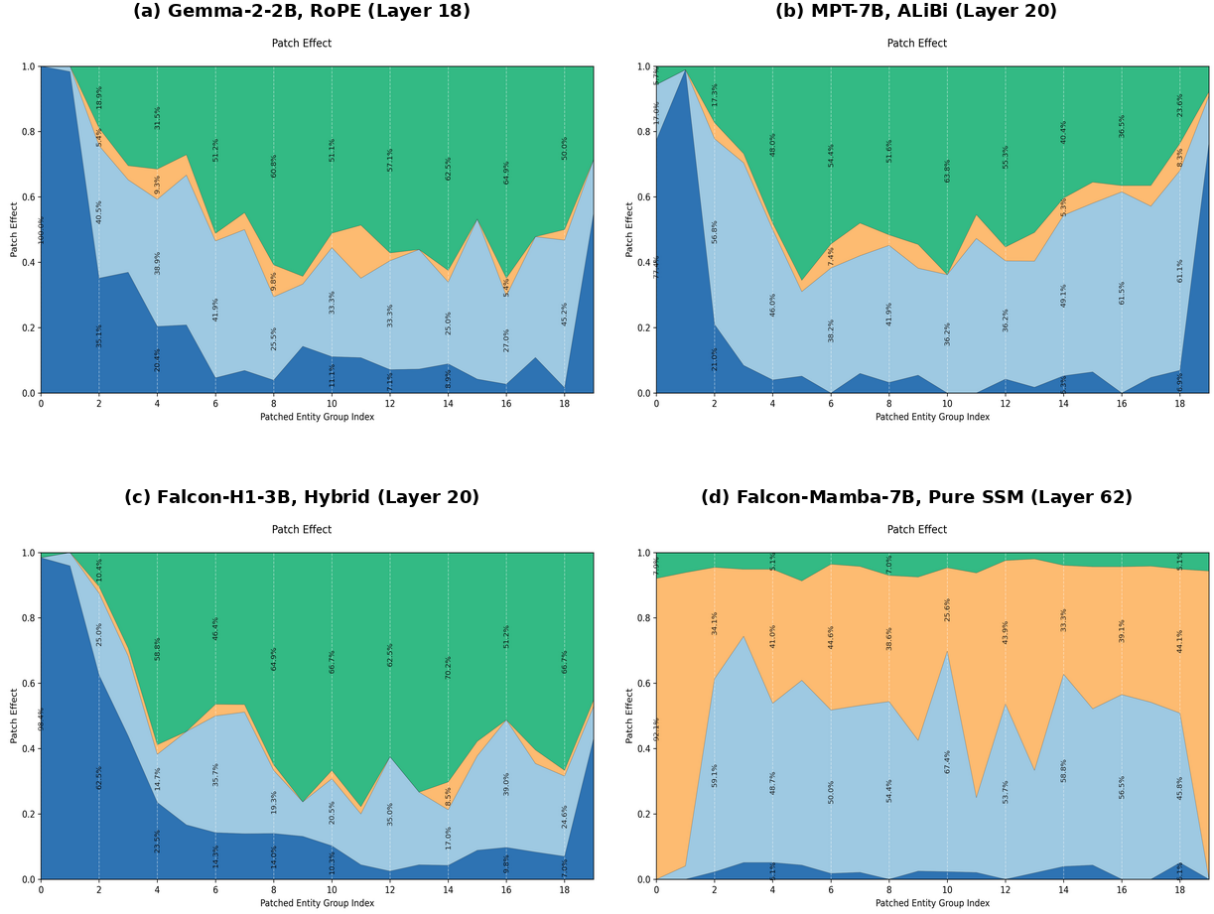


Figure 3: Binding effect by entity group position at a transition layer. (a–c) Attention-based architectures (RoPE, ALiBi, Hybrid) all show the same pattern: the first entity is bound positionally, with a transition to reflexive binding for later entities. (d) Falcon-Mamba-7B shows a primacy-recency effect with weak binding for middle entities, lacking the structured position dependence seen in attention models.

tity binding information in hybrid architectures. Component patching in Falcon-H1 shows that the attention component accounts for nearly all binding signal, while the Mamba component contributes negligibly. In Zamba2, binding transitions occur exclusively at hybrid layers—the only layers containing attention. This pattern holds despite the two architectures combining attention and Mamba in fundamentally different ways (parallel in Falcon-H1, sequential in Zamba2). One possible explanation is that attention’s pairwise token interactions enable direct entity-attribute retrieval, while Mamba’s sequential state update may be less suited for this task.

Binding mechanisms are consistent across attention architectures. All three binding mechanisms—positional, lexical, and reflexive—are present across every attention-based architecture we studied (RoPE, ALiBi, Hybrid), including the same entity-position dependence where first

entities are bound positionally and middle entities transition to reflexive binding. This suggests these mechanisms may reflect a general property of attention rather than an artifact of a specific positional encoding.

Pure Mamba shows a different binding pattern.

Falcon-Mamba-7B achieves 100% task accuracy yet shows a different binding pattern: no intermediate positional or lexical phases, only ~50% reflexive binding at the final layer, and a persistent ~40% mixed category that our classification framework cannot decompose. The primacy-recency effect further suggests that Mamba’s recurrent state may struggle to maintain uniform binding across entity positions. Whether the mixed signal reflects a genuinely different binding mechanism that our framework does not capture, or a noisier implementation of similar mechanisms, remains an open question.

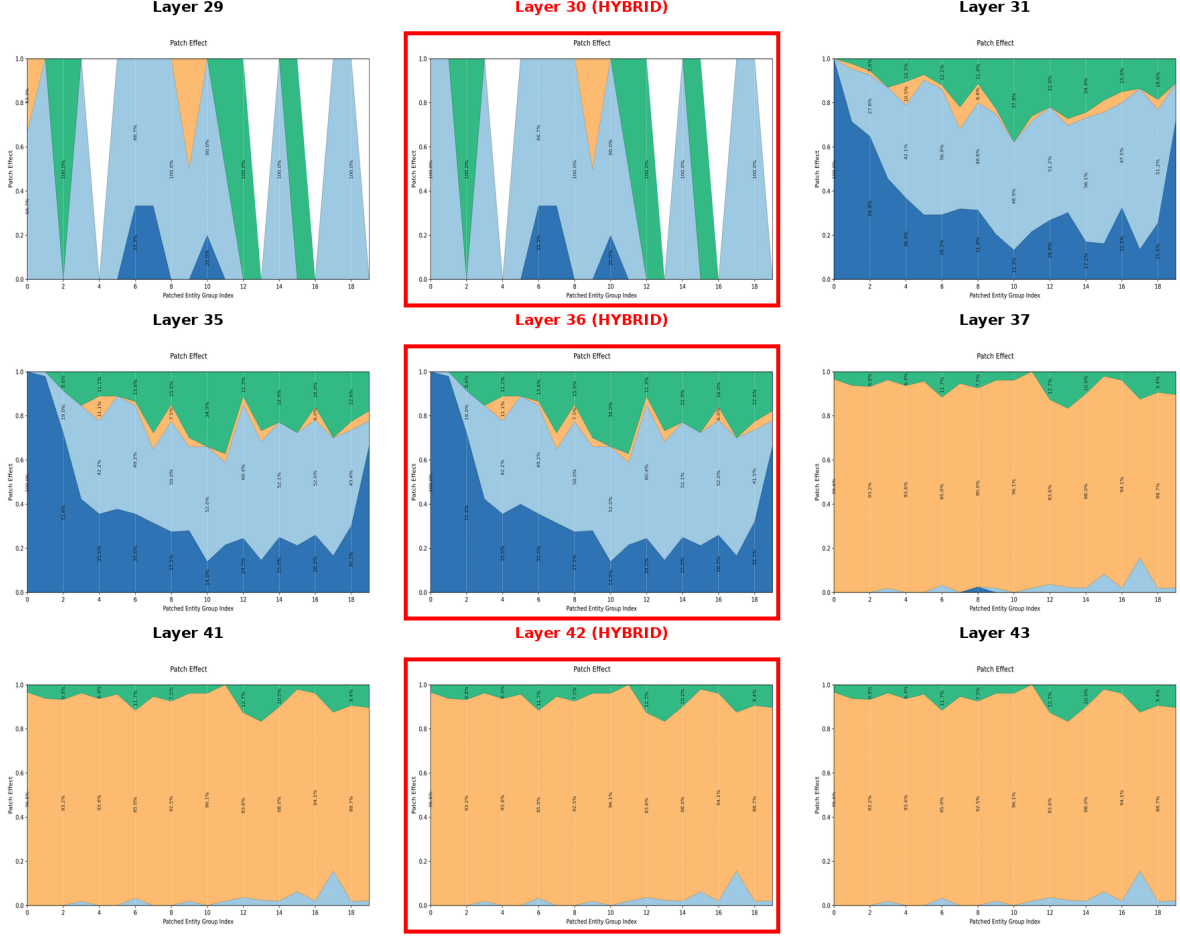


Figure 4: Zamba2-2.7B residual stream patching at consecutive layers around three hybrid positions (layers 30, 36, 42; marked with red borders). Each row shows the pure Mamba layer before, the hybrid layer, and the pure Mamba layer after. Binding transitions occur only at hybrid (attention-containing) layers, while pure Mamba layers show no change.

Limitations. Our study is limited to models up to 7B parameters due to university hardware constraints (NVIDIA Titan and RTX 2080 GPUs), and few models with alternative positional encodings (ALiBi, Absolute) or pure Mamba architectures achieve sufficient accuracy on the entity binding task at this scale. Extending to larger models or more diverse architectures may reveal additional patterns. The multi-layer component patching approach replaces a component at all layers simultaneously, which may not perfectly isolate its causal role if components interact nonlinearly across layers. Additionally, our classification framework—designed for attention-based models—may not fully capture Mamba’s binding strategy, as suggested by the high proportion of mixed effects.

Future work. Several directions remain open: (1) identifying the specific attention heads responsible for each binding mechanism, (2) developing

finer-grained classification methods to understand Mamba’s mixed binding signal, and (3) investigating how binding mechanisms develop during training. (4) identifying what property of ALiBi models enables scaling robustness at 100 entities.

Overall, our findings indicate that the three binding mechanisms identified by Gur-Arieh et al. (2025) generalize beyond RoPE to ALiBi and hybrid architectures, and that attention plays a central role in entity binding even when combined with state-space layers. Pure Mamba models can perform the binding task but appear to use a qualitatively different strategy that current classification frameworks do not fully capture. Extending interpretability methods to hybrid and recurrent models requires adapting both the experimental methodology and the analysis framework to account for these architectural differences.

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A AI Usage Disclosure and Reflection

We used Claude (Anthropic) and GitHub Copilot throughout this project.

How we used AI. We used AI tools for: (1) writing and debugging experiment code (activation patching hooks, multi-layer component patching, shared module handling for Zamba2), (2) analyzing experimental results (reading CSV files, computing distributions, generating figures), (3) drafting and iterating on the paper text (structure, wording, LaTeX formatting), and (4) reviewing the paper for factual accuracy and consistency.

What helped. AI was most helpful for implementation tasks—writing hooks for unfamiliar model architectures (Zamba2’s shared transformer blocks, Falcon-H1’s parallel components) and generating publication-quality figures from raw experimental data. It also helped identify factual errors during paper writing (e.g., verifying which positional encoding each model uses).

What hindered. AI occasionally made incorrect claims about model architectures that required manual verification. It also tended to overclaim results (e.g., using “universal” or “fundamentally” when the evidence supported softer claims). Multiple rounds of review were needed to ensure accuracy and appropriate hedging.